POTPRESS · PROJECT PLAN

PotPress project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potpress		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Concept selection decided (D1, option A); D6 threshold 5.4 t after D9 (Amish, 2026-10-25)
0.2	2026-10-23	Amish Chadha	Updated for the STEM check fixes (PPR-DDR-006)
0.1	2026-10-23	Amish Chadha	First plan (kit 1.8 upgrade); stage P3 at the portfolio cap

PotPress project plan

PotPress is at stage P3 (TRL 3, design on paper), the portfolio's stage cap. The next gate is P4, a first prototype with a partner factory, which stays closed until Amish opens it. The decision requested now is D1 in PPR-DEC-001 (D6 waits for pressing trials). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Science check	Design lead	PPR-STM-001	Pass; four fixes made 2026-10-23 (PPR-DDR-006)
Concept selection	Amish	PPR-OPT-001, PPR-DDR-005	Decided by Amish, 2026-10-25: option A (D1)
Requirements	Design lead	PPR-URS-001, PPR-PRS-001, PPR-RTM-001	Drafted; one Medium gap (UR-013)
Structure and guarding	Design lead	PPR-CAL-001, drawings, model	Met on paper; guard check unsigned
Build plan	Design lead	PPR-BLD-001	Done (TRL 3)
Regulatory, feasibility, risk, market	Design lead	PPR-REG-001, PPR-FEA-001, PPR-RSK-001, PPR-MKT-001	First pass done
Partner and co-design	Amish	Partner chosen; co-design notes	Not started (portfolio rule)

Resources needed

Table 2. Resources (estimates for the P4 prototype; on hold).

Resource	Deliverable	Effort	Estimated cost (USD)
Steel, jack, hardware, guards, lifting set	Frame and guarded press	About 3 days of fabrication	About 805 (BOM; the $66 lifting set can be hired)
Pattern printing and casting	Two molds	About 2 days printing, foundry lead time	About 391 (BOM), plus the melt check (to assess)
QC rack	Four-station rack	About 1 day	About 83 (BOM)
Competent person	Signed ISO 13857 desk check	Half a day	To assess
Partner factory time	Pressing trials and co-design	About 2 weeks	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Idea	Problem, STEM check, concept options, regulatory, feasibility, plan	Done (kit 1.8 documents added 2026-10-23)
P1 Concept	User and product requirements, risk, market	Done
P2 Design	Calculations, traceability matrix	Done
P3 Detailed design	Constructable model, drawings, build plan, decisions register, design FMEA	Done on paper; STEM check pass
P4 Prototype	Partner chosen, guard check signed, prototype built and tested	On hold (stage cap)
P5 Field trial	Pressing trials at a partner factory	On hold
P6 Release	Released drawings and precis at v1.0	On hold

Next steps and decision requested

1. Amish decides D1 (keep option A).
2. Render the photoreal views and storefront cards on Amish's Mac from the updated appearance model.
3. Propose a product requirement for local spares (UR-013).
4. At TRL 4, with a partner: shrinkage test bar, foundry melt check, EN 1005-3 handle assessment, and D6 if trials need more than 5.4 t.

References

- Ceramics Manufacturing Working Group. *Best Practice Recommendations for Local Manufacturing of Ceramic Pot Filters for Household Water Treatment.* 2011.
- Henry, M., S. Maley, and K. Mehta. "Designing a Low-Cost Ceramic Water Filter Press." *International Journal for Service Learning in Engineering* (2013). https://ojs.library.queensu.ca/index.php/ijsle/article/view/4532.
- Rayner, J. *Current Practices in Manufacturing Locally-Made Ceramic Pot Filters for Water Treatment in Developing Countries.* Loughborough: WEDC, 2009.
- Soppe, A. I. A., S. G. J. Heijman, et al. "Critical Parameters in the Production of Ceramic Pot Filters." *Journal of Water and Health* 13, no. 2 (2015).
- World Health Organization. *Results of Round I of the WHO International Scheme to Evaluate Household Water Treatment Technologies.* 2016.
- Further sources: PPR-PRB-001, PPR-CAL-001, PPR-STM-001, PPR-REG-001 and PPR-MKT-001.